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1 WELCOME TO COURSE :-

→ HARD WORK + CONSISTENCY

2. COURSE INTRODUCTION:

1. Topic list :-
2. Path To placement

Step 1 : Learn a language
Step 2 : Learn Data Structure & Algorithms
Step 3 (live) : Projects, Theory subjects, Resume, Refferals, Applying to companies, Interview preparation.

→ Harder ↑ = Luckier ↑

3. What will you learn :-

- Learn complete Java.
- Learn complete Data structure & algorithms
- Questions practice on all topics
- Guidence about next step.

4. Prerequisties :-

→ A laptop + Internet
→ Install Java + code error
→ Notebook + Pen

Error
Exception
—
—

Google
↓
Stack
Overflow

3. PRE-REQUISITES & Installation Resources

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- what to install :-

  ① visual Studio code
  ② Java Development Kit (JDK)

→ Solve your own problem.

4. Flow - Charts - Pseudocode :-

□ What are flow charts ? (logical Building)

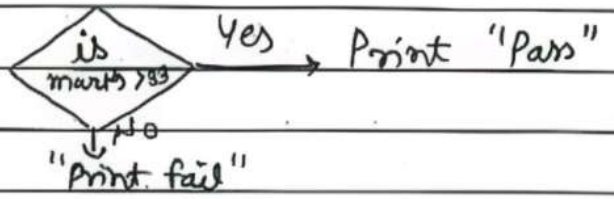
→ Diagram to represent solution of problems.

A problem → ① Small parts
↓
② Logically Arrange

① Tea :-

Start
↓
Bowl the water
↓
Gas store on
↓
Sugar, Tea, Milk
↓
Boil
↓
Exit

Flow chart Components:-

- Start / Exit → (Start)
- Input / Output → /Read N/ /Print "hi"/
- Process → [Name = "Vishal"]
- ~~Des.~~ Decision → 

Sum of 2 Numbers:-

Input:-

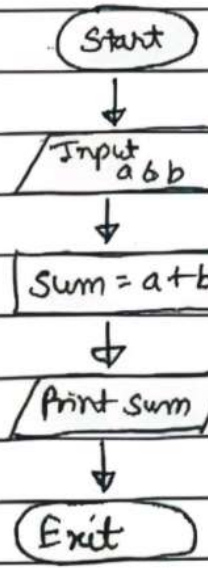
First number, a

Second number, b

Output:-

Sum of a & b

Flowchart



Pseudocode

1. Start
2. Input numbers a & b
3. calculate sum = a+b
4. Print sum
5. Exit

Calculate Sample Interest:-

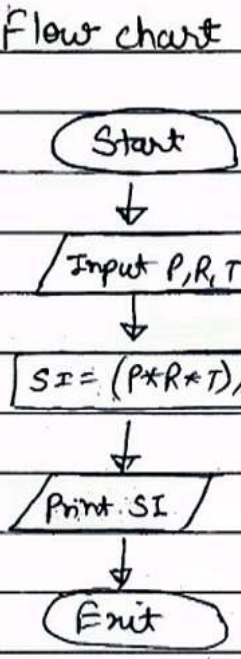
Input:-

Principal, P
Rate, R
Time, T

Output:-

$SI = PRT/100$

Flow chart



Pseudocode

1. Start
2. Input, Principal (P) Rate (R), & Time (T)
3. Calculate

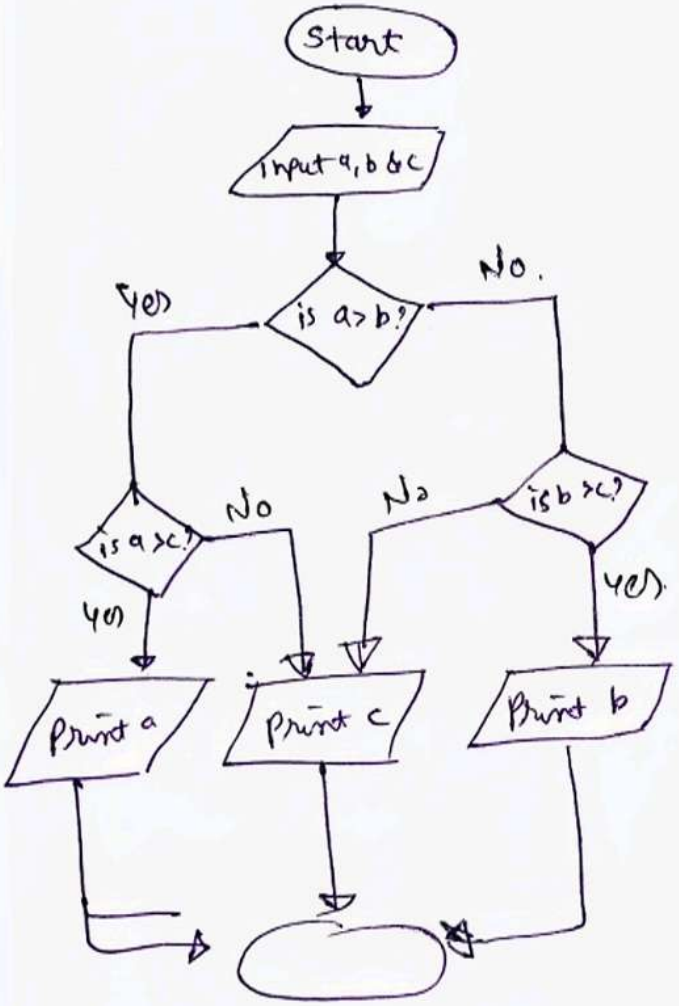
   $S = (P*R*T)/100$
4. Print SI
5. Exit

- Max of 3 Numbers

Input : 3 numbers (a, b, c)

Output : Max of 3

flowchart



Pseudocode.

1. Start
2. Input a, b & c
3. if a > b do
   if a > c do
   Print a.
   else
   Print c
   else
   if b > c do
   Print b
   else
   Print c
4. Exit

- find if a number is Prime or Not

Input.

number, n

Output :-

Prime or Not prime.

Non prime (n)

2 ⟷ n-1

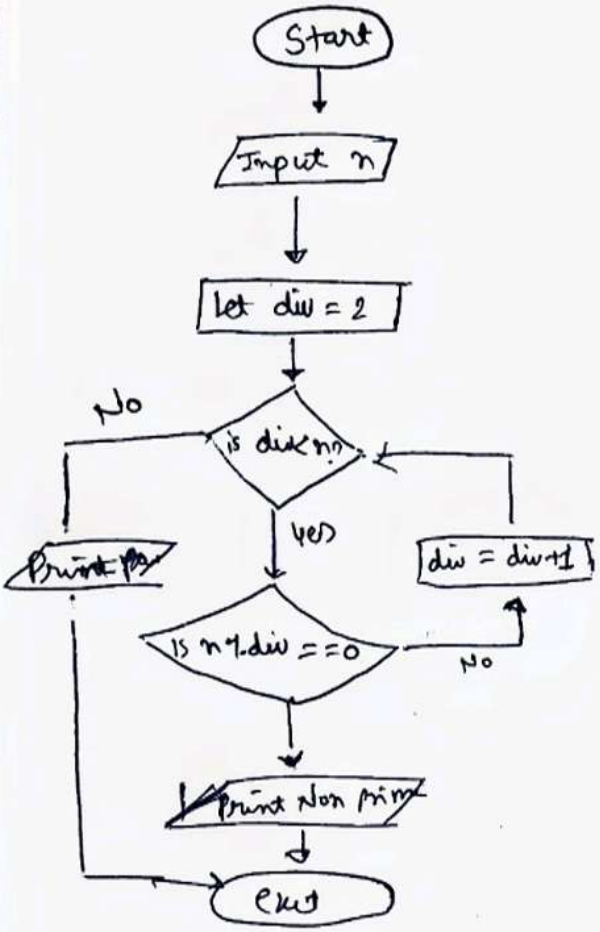
div n/div → rem = 0

6

2 5

(2, 3)

Flowchart



Pseudo code.

1. Start
2. Input n
3. ~~while~~ let div = 2
4. while div < n do
   if N % div == 0 do
      Print "Not Prime"
      Exit
   else
      div = div + 1
5. Print prime No.
6. Exit

- Sum of first N natural Numbers.

n = 4

Sum = 1+2+3+4 = 10

Input : numbers, n

output : sum of 1st Natural No.

logic :-

Sum = 1 to n (Value)

{Val <= n}

Val = 1 2 3 4

Sum = sum + Value

Sum = 0 Val = 1

Sum = 0+1 = 1

Sum = 1+2 = 3

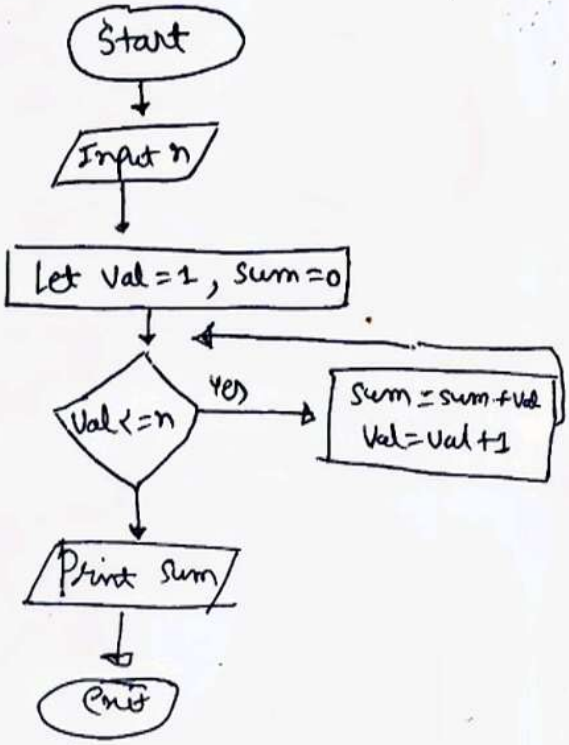
Sum = 3+4 = 10

Sum = 10+5 = 15

Pseudo code

1. Start
2. Input n
3. Let value = 1 &
   sum = 0
4. while val <= n do
   sum = sum + val
   val = val + 1
5. Print sum
6. exit

Flowchart

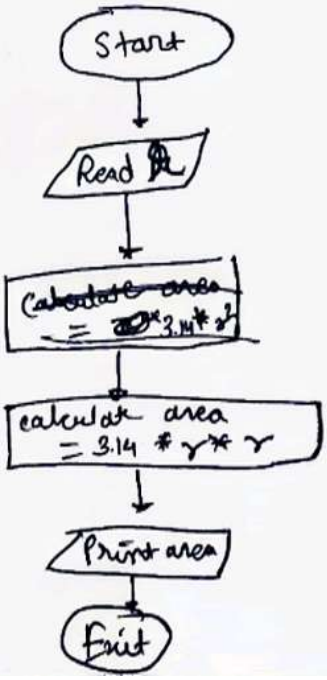


▷ Practise Questions

1. Flowchart to calculate the area of a circle.

Input : r
Output : area

Flowchart



Pseudocode

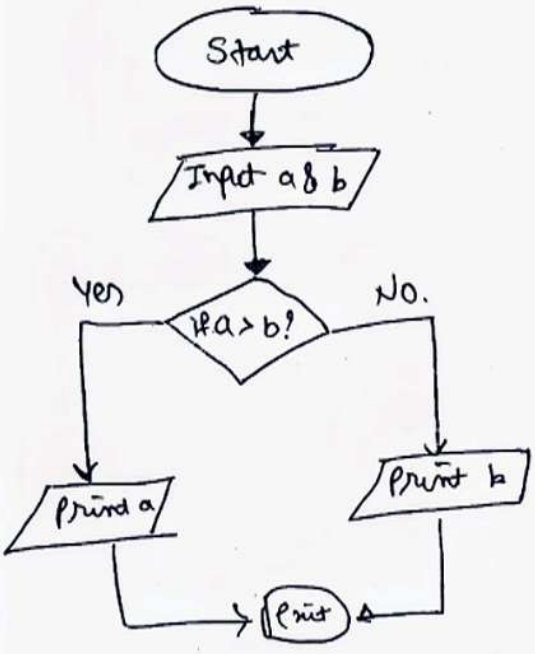
1. Start
2. Read r
3. Input r
4. Calculate area
   a = 3.14 * r * r
5. Print area
6. Exit

② Flowchart to find the Greatest from 2 numbers

Input : numbers , a & b

Output : Greatest No.

Flow chart

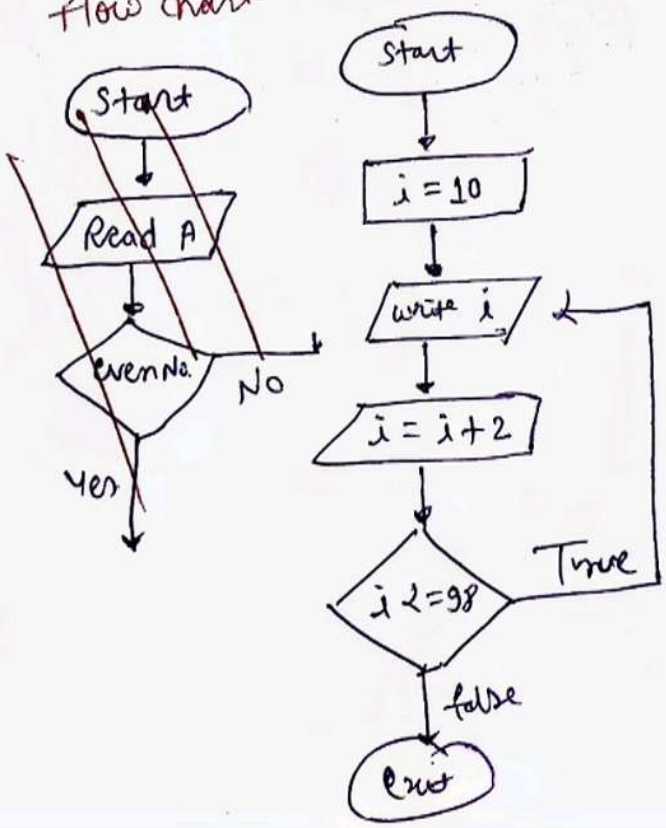


Pseudocode

1. Start
2. Input, a & b
3. if a > b do
   Print a.
   else
   print b
4. Exit

③ Flowchart & Pseudocode to print the even numbers ~~b/w 9 to~~ b/w 9 and 100.

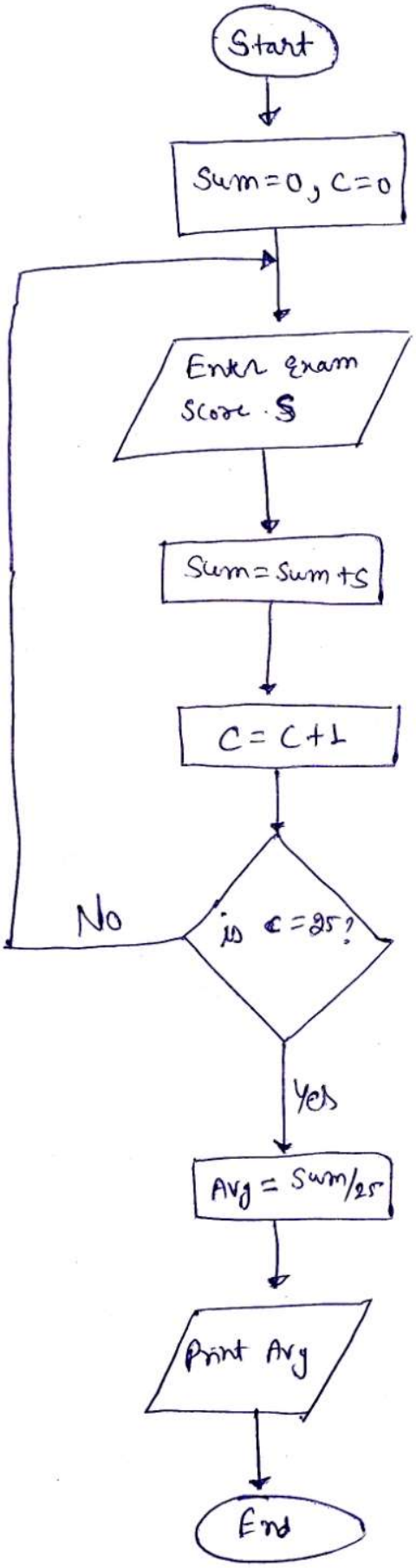
Flow chart



Pseudocode

1. Start
2. Start with i = 10
3. Write i
4. Than i = i+2
5. if i <= 98 do
   Print i
   else
   Exit
6. Exit.

(4) Flowchart for calculating the average from 25 exam scores.



Pseudocode

(1) Start

(2) SEt Sum = 0
set C = 0

(3) while C< 25 then do,
input S
Sum = Sum + S
C = C+1;
end of while

(4) Avg = sum /25

(5) Print Avg

(6) Ent